

INTRODUCTION TO APPLIED NUMERICAL METHODS

Curve Fitting: Lagrange Interpolating Polynomials

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NEWTON POLYNOMIAL INTERPOLATION

nth-Order Polynomial Interpolation

$$f_n(x) = b_0 + b_1(x - x_0) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

- (n+1) data points are required

$$[x_0, f(x_0)], [x_1, f(x_1)], \dots, [x_n, f(x_n)]$$



Coefficients to be evaluated:

$f[]$: finite divided difference

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

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$$b_n = f[x_n, x_{n-1}, \dots, x_1, x_0]$$

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j}$$

First finite divided difference

$$f[x_i, x_j, x_k] = \frac{f[x_i, x_j] - f[x_j, x_k]}{x_i - x_k}$$

Second finite divided difference

$$f[x_n, x_{n-1}, \dots, x_1, x_0] = \frac{f[x_n, x_{n-1}, \dots, x_1] - f[x_{n-1}, x_{n-2}, \dots, x_0]}{x_n - x_0}$$

nth finite divided difference

LAGRANGE INTERPOLATING POLYNOMIAL

- A reformulation of the Newton polynomial without computing divided differences

$$f_n(x) = \sum_{i=0}^n L_i(x) f(x_i)$$

Known value @ x_i

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

“product of”

$$L_1(x) = \frac{(x - x_0)(x - x_2)(x - x_3)(\cdots)(x - x_n)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)(\cdots)(x_1 - x_n)}$$

(n+1) data points available



$L_i(x)$ will be 1 at $x = x_i$ and 0 at all the other sample points x

$$f(x) = \frac{(x - x_1)(x - x_2)(x - x_3)(\cdots)(x - x_n)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)(\cdots)(x_0 - x_n)} f(x_0) + \frac{(x - x_0)(x - x_2)(x - x_3)(\cdots)(x - x_n)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)(\cdots)(x_1 - x_n)} f(x_1) + \dots + \frac{(x - x_0)(x - x_1)(x - x_2)(\cdots)(x - x_{n-1})}{(x_n - x_0)(x_n - x_2)(x_n - x_3)(\cdots)(x_n - x_{n-1})} f(x_n)$$

Each product $L_i(x)f(x_i)$ takes on the value of $f(x_i)$ at the sample point x_i

LAGRANGE INTERPOLATING POLYNOMIAL

Linear Lagrange Interpolating Polynomial (n = 1)

$$f(x) = \frac{(x - x_1)}{(x_0 - x_1)} f(x_0) + \frac{(x - x_0)}{(x_1 - x_0)} f(x_1) \quad \text{2 data points available}$$
$$= \frac{f(x_1)(x - x_0) - f(x_0)(x - x_1)}{(x_1 - x_0)} = \frac{f(x_1) - f(x_0)}{(x_1 - x_0)} x + \frac{x_1 f(x_0) - x_0 f(x_1)}{(x_1 - x_0)}$$

2nd-order Lagrange Interpolating Polynomial (n = 3)

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} f(x_1)$$
$$+ \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} f(x_2)$$

AN EXAMPLE

Use a Lagrange interpolating polynomial of the second order to evaluate $\ln 2$ using the data given below

(True value of $\ln(2) = 0.6931472$)

$$x_0 = 1 \quad f(x_0) = 0$$

$$x_1 = 4 \quad f(x_1) = 1.386294$$

$$x_2 = 6 \quad f(x_2) = 1.791760$$

Solution:

The second order Lagrange polynomial :

$$f_2(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}f(x_0) + \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}f(x_1) + \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}f(x_2)$$



$$f_2(x) = \frac{(x-4)(x-6)}{(1-4)(1-6)}0 + \frac{(x-1)(x-6)}{(4-1)(4-6)}1.386294 + \frac{(x-1)(x-4)}{(6-1)(6-4)}1.791760 \quad @ x = 2$$

$$f_2(2) = \frac{(2-4)(2-6)}{(1-4)(1-6)}0 + \frac{(2-1)(2-6)}{(4-1)(4-6)}1.386294 + \frac{(2-1)(2-4)}{(6-1)(6-4)}1.791760 = 0.5658444$$

AN EXAMPLE

Solution:

The second order Lagrange polynomial :

$$x_0 = 1 \quad f(x_0) = 0$$

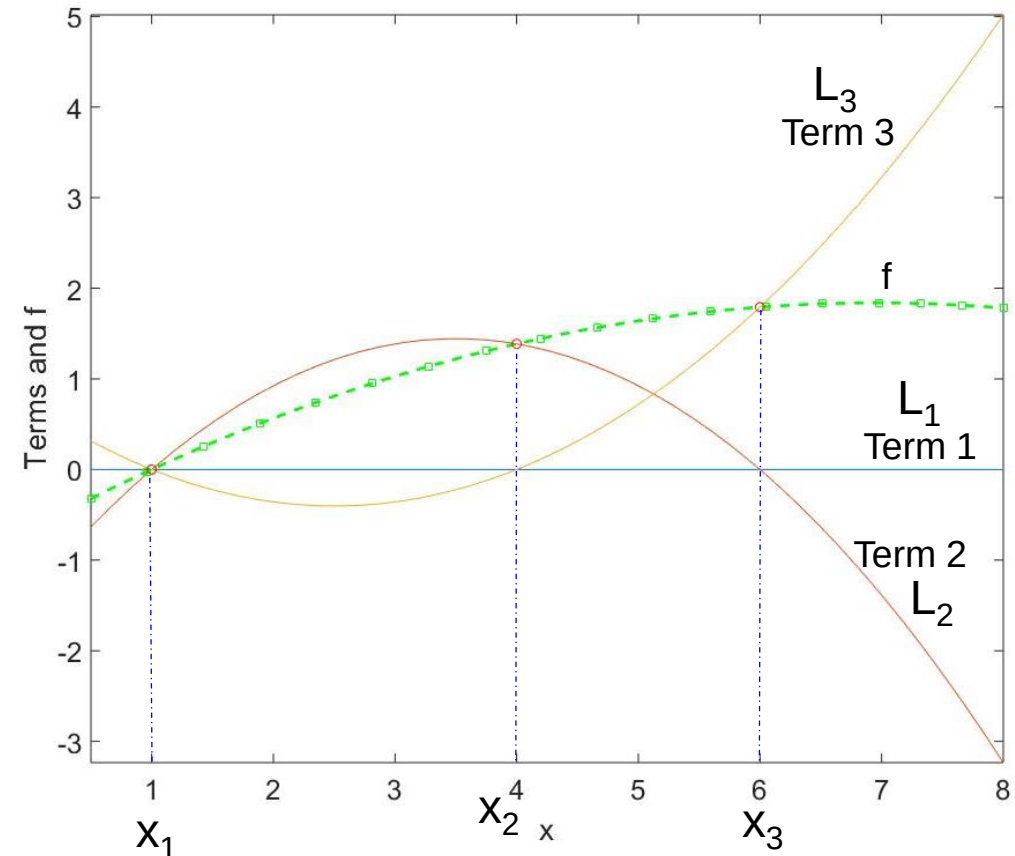
$$x_1 = 4 \quad f(x_1) = 1.386294$$

$$x_2 = 6 \quad f(x_2) = 1.791760$$

$$f_2(x) = \frac{(x-4)(x-6)}{(1-4)(1-6)} 0 + \frac{(x-1)(x-6)}{(4-1)(4-6)} 1.386294 + \frac{(x-1)(x-4)}{(6-1)(6-4)} 1.791760$$

$L_i(x)$ will be 1 at $x = x_i$ and 0 at all the other sample points x

Each product $L_i(x)f(x_i)$ takes on the value of $f(x_i)$ at the sample point x_i



THE DANGER OF USING AN INTERPOLATING POLYNOMIAL FOR EXTRAPOLATION

